# Appendix A

## Social Media Questionnaire

*Note.* The questionnaire was administered in Spanish via Microsoft Forms on school computers; an English translation is provided here. The asterisk (*) indicates a compulsory question. The questionnaire integrates the ERA-RSI scale, the Pittsburgh Sleep Quality Index, the Rosenberg Self-Esteem Scale and the EIDAN scale. Questions 12–37, 38–63 and 64–88 are displayed conditionally depending on the most-used social media platform (TikTok, Instagram or others), and questions 117–146 are displayed to those who report not using social media.

### Initial data

1. Before starting, indicate your age in order to read the appropriate consent statement: *

   *11 years old / 12 years old / 13 years old / 14 years old / 15 years old / 16 years old / 17 years old / 18 years old / 19 years old*

2. Also indicate your school year: *

   *1st ESO / 2nd ESO / 3rd ESO / 4th ESO / 1st Bachillerato / 2nd Bachillerato*

### Information and informed consent

Depending on the age indicated in question 1, the form displays one of the two information sheets below, whose text is reproduced in full (names of persons and institutions have been anonymized, the decorative icons of the original form have been omitted, and the text has been translated from the original Spanish).

### Information and Consent to Participate in this Study on Social Media (version for minors)

What is this study about? We want to understand how the use of TikTok and other Social Media affects adolescents. We will investigate how algorithms (the ones that

decide which videos appear to you) can influence things such as your mood, sleep quality, self-esteem or anxiety. Our goal is to see whether these algorithms can be improved to make social media safer and healthier for young people. Your participation is key: with your answers, we will be able to find solutions so that TikTok and other Social Media do not negatively affect the health of adolescents.

Who is carrying out this study? This study is carried out by a team of researchers from **[ANONYMIZED]** and **[ANONYMIZED]**. **[ANONYMIZED]** – Principal investigator, professor of Economics at **[ANONYMIZED]**. Email: **[ANONYMIZED]**. **[ANONYMIZED]** – Researcher and bioethics expert at **[ANONYMIZED]**. Email: **[ANONYMIZED]**.

If you have any doubts or need more information about how to withdraw your participation, or information about the confidentiality or anonymization of the data, you can write to them at any time.

What do I have to do? You will simply answer a questionnaire about how you use TikTok or other Social Media and how you feel in your day-to-day life. It will be done on a school computer, and there are no right or wrong answers.

– You can answer all the questions, only some, or none.

– If at any point you decide you do not want to continue, you can leave without any problem.

– Nobody (neither the researchers, nor the teachers, nor your parents) will force you to continue if you do not want to.

What information will be collected? The questions will deal with topics such as:

– How much you use TikTok or other Social Media and how it makes you feel.

– Whether it affects your sleep or how you relate to others.

– Whether you sometimes feel anxiety, stress or pressure because of Social Media.

Is it safe? What happens to my data? Your privacy is 100% protected. This means that: Your answers are anonymous; nobody will know what you answered. Your data will not be shared with anyone without your permission. Your answers will only be used for this study and, if they are used for other studies, they will always be anonymous. This research project very strictly follows the European data protection rules (the famous GDPR) and also the rules of the Ethics Committee of **[ANONYMIZED]**. The researchers will comply with the Data Protection Law (European Regulation 679/2016 and Organic Law 3/2018) to guarantee your safety. Special measures have been put in place so that all the information collected is secure, confidential and used only as necessary. These measures are based on storing few data, only for the time strictly needed, and protecting them from unauthorized access, as required by Article 5 of the GDPR.

The main measures we are using are:

– Secure tools: We use programs that comply with GDPR rules to collect, store and process personal data. In addition, all information travels in a protected way thanks to encryption systems (like a very strong digital padlock).

– Secure databases: The data are stored in the cloud, within the Microsoft 365 system used by **[ANONYMIZED]**, which is very secure and complies with the GDPR. Only the researchers working on the project will have access, and they will have different permissions depending on their role.

– Extra data protection: 1. Pseudonymization: We will not use participants' email addresses directly. Instead, they will be replaced by randomly generated secret codes that only the principal investigators will be able to decipher. 2. Hashing: The codes will be further protected using a special technique (SHA-256) which means that, even if someone obtained them, they could not know who they belong to without the secret key. 3. Encryption: The list linking email addresses with

codes will be encrypted with an even stronger system (AES-256) and only a few researchers will have access.

– Continuous review: The terms and conditions of all the programs we use will be reviewed periodically to make sure they remain secure.

– Ongoing advice: We work together with the Data Protection Officer of **[ANONYMIZED]** (whom you can contact at any time at **[ANONYMIZED]**) and with ethics experts to always maintain a high level of security adapted to the risks.

– Limited storage time: We will only keep the data for the duration of the project. Afterwards, the data will be permanently and securely deleted so that nobody can recover them.

In short: we take participants' privacy very seriously. We want all their data to be protected and the project to fully respect their rights.

How can I withdraw my consent? If at any point you decide that you no longer want to participate in the study, you can notify the principal investigator: **[ANONYMIZED]** – Email: **[ANONYMIZED]**. You will not have to give any explanation and there will be no consequences. You can also ask us to delete any of your data that has already been collected.

How long will it take? The study will last about 6 months, but you will only have to answer the questionnaire twice, and it only takes a few minutes.

If I accept, what do I do? If you agree to participate, accept the consent below. If you have any doubts, ask the person in charge present in your room before deciding.

3. YOU HAVE NOW READ THE CONSENT STATEMENT *

*I HAVE READ THE CONSENT STATEMENT AND I AGREE TO PARTICIPATE IN THE STUDY / I HAVE READ THE CONSENT STATEMENT AND I REFUSE TO PARTICIPATE IN THE STUDY*

**Information and Consent to Participate in this Study on Social Media (formal version)**

NAME OF THE STUDY/PROJECT: Bioethical implications of minors' continued access to the internet: algorithmic ethics of artificial intelligence-based systems.

PRINCIPAL INVESTIGATOR: 1. Name: **[ANONYMIZED]**. 2. Affiliation: **[ANONYMIZED]**. 3. Email address: **[ANONYMIZED]**.

We thank you for your interest in this study. In order to participate, your consent is required, so we ask you to take the necessary time to read the following information carefully, and to ask the researcher whose contact details appear above as many questions as you consider necessary, so that you can consider yourself truly informed. Please be aware that, in any case, your decision to participate is entirely voluntary and that, even if you have decided to participate, you may withdraw your consent at any time, without any consequence for you.

What is this study and what does it pursue? This study/project aims to analyse the use of Social Media among young Spaniards, especially the use of the social network TikTok, but not only. The study aims to identify the impact that the algorithms of TikTok or other Social Media, which are developed with Artificial Intelligence, may have on the well-being of underage students. Through the study of the ways TikTok or other Social Media are used and their possible effects on personal well-being, from sleep quality to self-esteem or anxiety, the study hopes to identify the possible problems of these algorithms, in order finally to propose alternative ways of programming and marketing social media without risk for minors. For this reason, your participation in the questionnaire is essential, since only with your answers to the questions will the research team be able to find evidence of the possibly inadequate algorithms of TikTok or other

Social Media that harm students like you. Thus, with your participation you will be facilitating the collection of scientific data to create healthier environments for young Spaniards in the digital world, where social media do not harm their health but rather foster positive relationships with friends and family.

– **[ANONYMIZED]** (PI), Full Professor at **[ANONYMIZED]**, PhD in Economics. **[ANONYMIZED]**
– **[ANONYMIZED]**, Full Professor at **[ANONYMIZED]**, PhD in Psychology. **[ANONYMIZED]**
– **[ANONYMIZED]**, Associate Professor at **[ANONYMIZED]**, PhD in Economics. **[ANONYMIZED]**
– **[ANONYMIZED]**, PhD candidate, **[ANONYMIZED]**, Programme in Health, Well-being and Bioethics. **[ANONYMIZED]**

How long will the study last? The expected duration of this study is 6 months, with different interventions from obtaining consent (September 2025), the initial data collection through questionnaires (October 2025), the delivery of training workshops (October–November 2025), the second administration of the questionnaire (December 2025) and the collection of data from the training workshops (January–February 2026).

How will the study be carried out? What will my participation consist of and what risks does it entail? Your participation will consist of answering various questions in a single questionnaire related to your personality and the use of TikTok or other Social Media. This questionnaire measures the variables of sleep quality, physical health, anxiety and psychological well-being. This questionnaire will be completed once using a school computer, with the questions presented in digital format. You may answer all, none or some of the questions, and you may even withdraw before or during the test

without any obligation or consequence for you, either from the researchers, your parents or the teachers.

What data will be collected for this study? The data processed in the joint questionnaire, based on the following scales, are listed below:

- 1. Adolescent Social Media and Internet Addiction Risk Scale (ERA-RSI): Links between symptoms and social media addiction. Social use of social media. Personal traits derived from the social use of social media. Irrational fear of not having one's mobile phone.
- 2. Pittsburgh Sleep Quality Index: Subjective sleep quality. Sleep latency. Sleep duration. Habitual sleep efficiency. Sleep disturbances. Use of sleep medication. Daytime dysfunction.
- 3. Rosenberg Self-Esteem Scale: Positive self-esteem references (worth, virtue, acceptance and integration). Negative self-esteem references (failure, shame, uselessness and badness).
- 4. Depression and Anxiety Interference Scale for Children and Adolescents (EIDAN): Relationships with siblings/family members. School homework or studying. Relationships with classmates. Daily activities. Relationships with parents. Exams. Relationships with friends. Activities with the family. Class participation. Making new friends.

How are the confidentiality and protection of my personal data guaranteed? This study involves the processing of personal data, so the researchers will guarantee confidentiality in their processing at all times, complying with personal data protection legislation, in particular European Regulation 679/2016 of 27 April, the General Data Protection Regulation, as well as Organic Law 3/2018 of 5 December on the Protection

of Personal Data and the Guarantee of Digital Rights. In application of data protection legislation, we inform you of the following:

DATA CONTROLLER. The data controller is **[ANONYMIZED]**.

CONSENT AND PURPOSE. Your personal data will be processed with your express consent and may only be used for additional and compatible research purposes after being anonymized. You may withdraw your consent at any time without any consequence for you.

DATA DISCLOSURE. The personal data collected will not be transferred without your express consent except in cases where there is a legal obligation to do so or after their anonymization, so that re-identification is not possible.

DATA RETENTION. Your personal data will be kept for the time strictly necessary for their possible anonymization (subsequently proceeding to their destruction).

EXERCISE OF RIGHTS. In application of your right to the protection of personal data, we inform you that you may at any time exercise your rights of access, rectification, erasure, restriction of processing, objection and the other rights recognized by the General Data Protection Regulation, as well as by Organic Law 3/2018 on the Protection of Personal Data and the guarantee of digital rights, by addressing your request to **[ANONYMIZED]**.

Likewise, and especially if you consider that you have not obtained full satisfaction in the exercise of your rights, we inform you that you may lodge a complaint with the national supervisory authority for data protection, by contacting the Spanish Data Protection Agency, C/ Jorge Juan, 6 – 28001 Madrid, or at www.aepd.es.

You can find more information about the protection of your personal data at **[ANONYMIZED]**. The Data Protection Officer (DPO) of **[ANONYMIZED]** can be contacted at **[ANONYMIZED]**.

4. THIS IS THE END OF THE INFORMATION SHEET FOR YOU TO CONSIDER WHETHER OR NOT YOU AGREE TO PARTICIPATE IN THE STUDY. We remind you that you may request any clarification or ask any question to make sure you have all the information you need to make your decision. If your decision is to participate, we ask you to complete and sign the following "Informed consent" sheet, signifying that you accept and consent to participate in the study after having received all the information. *

*I HAVE READ THE CONSENT STATEMENT AND I AGREE TO PARTICIPATE IN THE STUDY / I HAVE READ THE CONSENT STATEMENT AND I REFUSE TO PARTICIPATE IN THE STUDY*

**Sociodemographic and access data**

5. Indicate your gender: *

*Male / Female / Other*

6. Indicate the postcode of where you live:

*Open answer.*

7. Do you have a mobile phone? *

*Yes / No*

8. Do you have a computer? *

*Yes, and I use it on my own / Yes, and I share it with my family / No*

9. Do you have older or younger siblings? *

*Yes, I have older and younger siblings / Yes, I have older siblings / Yes, I have younger siblings / No*

10. Do you live with your parents or legal guardians? *

*Yes, with one of my parents or a legal guardian (choose this option if you spend some weeks with one and then with the other) / Yes, with both of my parents or legal guardians / No, with neither*

11. Which of these Social Media platforms do you use the most? *

*TikTok / Instagram / YouTube / Facebook / Other Social Media / I do not use Social Media*

*Note.* Question 11 determines the branch each participant sees: those who indicate TikTok answer questions 12–37; those who indicate Instagram, questions 38–63; those who indicate YouTube, Facebook or other social media, the generic block 64–88; and those who report not using social media go directly to the branch 117–146. Each adolescent therefore answers a single usage block (ERA-RSI) referring to their main platform. This design allows platform-specific analyses, testing whether the pathways under study differ across three architecturally distinct platforms: TikTok, Instagram and YouTube.

**TikTok use (ERA-RSI) — questions 12 to 37**

*Some people your age connect to the internet through mobile phones, such as iPhones, tablets, such as iPads, laptops or other devices such as game consoles. In this part of the questionnaire, answer how you use the internet or social media and what purpose you pursue. It is important that you answer all the statements below and that you do so honestly, marking the answer you consider appropriate according to the values indicated.*

12. How do you use the TikTok app? *

*Only on my mobile phone / Only on other devices (tablet, computer) / On my mobile phone and other devices (tablet, computer)*

13. Taking into account all the times I visit TikTok, the time I spend daily is: *

*Around 1 hour / About 2 hours / Between 3 and 4 hours / More than 4 hours*

14. I use TikTok during my study hours *

*Never / Sometimes / Quite often / Very frequently*

15. Right now I would feel angry if I had to stop using TikTok *

*Never / Sometimes / Quite often / Very frequently*

16. I access TikTok anywhere and at any time *

*Never / Sometimes / Quite often / Very frequently*

17. I believe that using TikTok has affected my grades *

*Never / Sometimes / Quite often / Very frequently*

18. I have lost hours of sleep because of using TikTok *

*Never / Sometimes / Quite often / Very frequently*

19. At home, I hide how much time I spend on TikTok *

*Never / Sometimes / Quite often / Very frequently*

20. If I do not have internet access I feel insecure *

*Never / Sometimes / Quite often / Very frequently*

21. I update my status *

*Never / Sometimes / Quite often / Very frequently*

22. I check my friends' profiles *

*Never / Sometimes / Quite often / Very frequently*

23. I use the chat *

*Never / Sometimes / Quite often / Very frequently*

24. I share photographs and/or videos *

*Never / Sometimes / Quite often / Very frequently*

25. We comment on photographs among friends *

*Never / Sometimes / Quite often / Very frequently*

26. I see what my contacts have been doing in the last few hours *

*Never / Sometimes / Quite often / Very frequently*

27. I find old friends *

*Never / Sometimes / Quite often / Very frequently*

28. The number of videos I have posted on TikTok is: *

*None / 1-15 / 16-50 / More than 50*

29. I make new friends *

*Never / Sometimes / Quite often / Very frequently*

30. I spend time on TikTok playing games *

*Never / Sometimes / Quite often / Very frequently*

31. I spend time on TikTok to join groups of interest *

*Never / Sometimes / Quite often / Very frequently*

32. I have a mobile phone and… I feel safer or more accompanied knowing that I can communicate with someone at any time *

*Never / Sometimes / Quite often / Very frequently*

33. I have a mobile phone and… I think it is safer to send a photograph by phone than to post it on other social media *

*Never / Sometimes / Quite often / Very frequently*

34. I have a mobile phone and… if people do not reply immediately to my messages I feel anxiety and distress *

*Never / Sometimes / Quite often / Very frequently*

35. I have a mobile phone and… I would be furious if my phone were taken away from me *

*Never / Sometimes / Quite often / Very frequently*

36. I have a mobile phone and… I need to know whether the recipient has read my message or not *

*Never / Sometimes / Quite often / Very frequently*

37. I have a mobile phone and… I get uneasy if nobody talks to me when we are online *

*Never / Sometimes / Quite often / Very frequently*

**Instagram use (ERA-RSI) — questions 38 to 63**

*Some people your age connect to the internet through mobile phones, such as iPhones, tablets, such as iPads, laptops or other devices such as game consoles. In this part of the questionnaire, answer how you use the internet or social media and what purpose you pursue. It is important that you answer all the statements below and that you do so honestly, marking the answer you consider appropriate according to the values indicated.*

38. How do you use the Instagram app? *

*Only on my mobile phone / Only on other devices (tablet, computer) / On my mobile phone and other devices (tablet, computer)*

39. Taking into account all the times I visit Instagram, the time I spend daily is: *

*Around 1 hour / About 2 hours / Between 3 and 4 hours / More than 4 hours*

40. I use Instagram during my study hours *

*Never / Sometimes / Quite often / Very frequently*

41. Right now I would feel angry if I had to stop using Instagram *

*Never / Sometimes / Quite often / Very frequently*

42. I access Instagram anywhere and at any time *

*Never / Sometimes / Quite often / Very frequently*

43. I believe that using Instagram has affected my grades *

*Never / Sometimes / Quite often / Very frequently*

44. I have lost hours of sleep because of using Instagram *

*Never / Sometimes / Quite often / Very frequently*

45. At home, I hide how much time I spend on Instagram *

*Never / Sometimes / Quite often / Very frequently*

46. If I do not have internet access I feel insecure *

*Never / Sometimes / Quite often / Very frequently*

47. I update my status *

*Never / Sometimes / Quite often / Very frequently*

48. I check my friends' profiles *

*Never / Sometimes / Quite often / Very frequently*

49. I use the chat *

*Never / Sometimes / Quite often / Very frequently*

50. I share photographs and/or videos *

*Never / Sometimes / Quite often / Very frequently*

51. We comment on photographs among friends *

*Never / Sometimes / Quite often / Very frequently*

52. I see what my contacts have been doing in the last few hours *

*Never / Sometimes / Quite often / Very frequently*

53. I find old friends *

*Never / Sometimes / Quite often / Very frequently*

54. The number of videos I have posted on Instagram is: *

*None / 1-15 / 16-50 / More than 50*

55. I make new friends *

*Never / Sometimes / Quite often / Very frequently*

56. I spend time on Instagram playing games *

*Never / Sometimes / Quite often / Very frequently*

57. I spend time on Instagram to join groups of interest *

*Never / Sometimes / Quite often / Very frequently*

58. I have a mobile phone and… I feel safer or more accompanied knowing that I can communicate with someone at any time *

*Never / Sometimes / Quite often / Very frequently*

59. I have a mobile phone and… I think it is safer to send a photograph by phone than to post it on other social media *

*Never / Sometimes / Quite often / Very frequently*

60. I have a mobile phone and… if people do not reply immediately to my messages I feel anxiety and distress *

*Never / Sometimes / Quite often / Very frequently*

61. I have a mobile phone and… I would be furious if my phone were taken away from me *

*Never / Sometimes / Quite often / Very frequently*

62. I have a mobile phone and… I need to know whether the recipient has read my message or not *

*Never / Sometimes / Quite often / Very frequently*

63. I have a mobile phone and… I get uneasy if nobody talks to me when we are online *

*Never / Sometimes / Quite often / Very frequently*

**Use of other Social Media (ERA-RSI) — questions 64 to 88**

*Some people your age connect to the internet through mobile phones, such as iPhones, tablets, such as iPads, laptops or other devices such as game consoles. In this part of the questionnaire, answer how you use the internet or social media and what purpose you*

*pursue. It is important that you answer all the statements below and that you do so honestly, marking the answer you consider appropriate according to the values indicated.*

64. Taking into account all the times I visit Social Media, the time I spend daily is: *

*Around 1 hour / About 2 hours / Between 3 and 4 hours / More than 4 hours*

65. I use Social Media during my study hours *

*Never / Sometimes / Quite often / Very frequently*

66. Right now I would feel angry if I had to stop using Social Media *

*Never / Sometimes / Quite often / Very frequently*

67. I access Social Media anywhere and at any time *

*Never / Sometimes / Quite often / Very frequently*

68. I believe that using Social Media has affected my grades *

*Never / Sometimes / Quite often / Very frequently*

69. I have lost hours of sleep because of using Social Media *

*Never / Sometimes / Quite often / Very frequently*

70. At home, I hide how much time I spend on Social Media *

*Never / Sometimes / Quite often / Very frequently*

71. If I do not have internet access I feel insecure *

*Never / Sometimes / Quite often / Very frequently*

72. I update my status on Social Media *

*Never / Sometimes / Quite often / Very frequently*

73. I check my friends' profiles on Social Media *

*Never / Sometimes / Quite often / Very frequently*

74. I use the chat on Social Media *

*Never / Sometimes / Quite often / Very frequently*

75. I share photographs and/or videos on Social Media *

*Never / Sometimes / Quite often / Very frequently*

76. We comment on photographs among friends on Social Media *

*Never / Sometimes / Quite often / Very frequently*

77. I see what my contacts have been doing in the last few hours on Social Media *

*Never / Sometimes / Quite often / Very frequently*

78. I find old friends on Social Media *

*Never / Sometimes / Quite often / Very frequently*

79. The number of videos I have posted on Social Media is: *

*None / 1-15 / 16-50 / More than 50*

80. I make new friends on Social Media *

*Never / Sometimes / Quite often / Very frequently*

81. I spend time on Social Media playing games *

*Never / Sometimes / Quite often / Very frequently*

82. I spend time on Social Media to join groups that interest me *

*Never / Sometimes / Quite often / Very frequently*

83. I have a mobile phone and… I feel safer or more accompanied knowing that I can communicate with someone at any time *

*Never / Sometimes / Quite often / Very frequently*

84. I have a mobile phone and… I think it is safer to send a photograph by phone than to post it on other social media *

*Never / Sometimes / Quite often / Very frequently*

85. I have a mobile phone and… if people do not reply immediately to my messages I feel anxiety and distress *

*Never / Sometimes / Quite often / Very frequently*

86. I have a mobile phone and… I would be furious if my phone were taken away from me *

*Never / Sometimes / Quite often / Very frequently*

87. I have a mobile phone and… I need to know whether the recipient has read my message or not *

*Never / Sometimes / Quite often / Very frequently*

88. I have a mobile phone and… I get uneasy if nobody talks to me when we are online *

*Never / Sometimes / Quite often / Very frequently*

**Sleep quality (Pittsburgh Index) — questions 89 to 106**

*The following questions refer to how you have slept during the past month. Try to answer as accurately as possible about your sleep during most of the days and nights of the past month. It is important that you answer all the questions below and that you do so honestly, marking the answer you consider appropriate.*

89. During the past month, what has usually been your bedtime? *

*9 to 10 / 10 to 11 / 11 to 12 / 12 to 1 / After 1*

90. During the past month, how long has it taken you to fall asleep at night? *

*Up to 15 minutes / 15 to 30 minutes / 30 to 60 minutes / More than 60 minutes*

91. During the past month, what time have you been getting up in the morning? *

*Before 5 / 5 to 6 / 6 to 7 / 7 to 8*

92. How many hours of actual sleep do you estimate you have had each night during the past month? *

*Less than 6 hours / 6 to 7 hours / 7 to 8 hours / 8 to 9 hours / 9 to 10 hours / More than 10 hours*

93. During the past month, how often have you had trouble sleeping… because you could not fall asleep within the first half hour? *

*Not during the past month / Less than once a week / Once or twice a week / Three or more times a week*

94. During the past month, how often have you had trouble sleeping… because you woke up during the night or in the early morning? *

*Not during the past month / Less than once a week / Once or twice a week / Three or more times a week*

95. During the past month, how often have you had trouble sleeping… because you had to get up to use the bathroom? *

*Not during the past month / Less than once a week / Once or twice a week / Three or more times a week*

96. During the past month, how often have you had trouble sleeping… because you could not breathe comfortably? *

*Not during the past month / Less than once a week / Once or twice a week / Three or more times a week*

97. During the past month, how often have you had trouble sleeping… because you coughed or snored loudly? *

*Not during the past month / Less than once a week / Once or twice a week / Three or more times a week*

98. During the past month, how often have you had trouble sleeping… because you felt cold? *

*Not during the past month / Less than once a week / Once or twice a week / Three or more times a week*

99. During the past month, how often have you had trouble sleeping… because you felt too hot? *

*Not during the past month / Less than once a week / Once or twice a week / Three or more times a week*

100. During the past month, how often have you had trouble sleeping… because you had nightmares or bad dreams? *

*Not during the past month / Less than once a week / Once or twice a week / Three or more times a week*

101. During the past month, how often have you had trouble sleeping… because you were in pain? *

*Not during the past month / Less than once a week / Once or twice a week / Three or more times a week*

102. During the past month, how often have you had trouble sleeping… for other reasons? (write them below and indicate their frequency) *

*Open answer.*

103. During the past month, how would you rate your sleep quality overall? *

*Fairly good / Good / Bad / Fairly bad*

104. During the past month, how often have you taken medicine (on your own or prescribed by a doctor) to help you sleep? *

*Not during the past month / Less than once a week / Once or twice a week / Three or more times a week*

105. During the past month, how often have you felt sleepy at school or while doing other activities? *

*Not during the past month / Less than once a week / Once or twice a week / Three or more times a week*

106. During the past month, has it been a problem for you to lack the motivation to do some school or other activities? *

*No problem at all / A very slight problem / A problem / A big problem*

**Self-esteem (Rosenberg Scale) — questions 107 to 116**

*The following questions refer to how you feel. It is important that you answer all the questions below and that you do so honestly, marking the answer you consider appropriate.*

107. I feel that I am a person of worth, at least on an equal plane with others *

*Strongly disagree / Disagree / Agree / Strongly agree*

108. All in all, I am inclined to feel that I am a failure *

*Strongly disagree / Disagree / Agree / Strongly agree*

109. I feel that I have a number of good qualities *

*Strongly disagree / Disagree / Agree / Strongly agree*

110. I am able to do things as well as most other people *

*Strongly disagree / Disagree / Agree / Strongly agree*

111. I feel I do not have much to be proud of *

*Strongly disagree / Disagree / Agree / Strongly agree*

112. I take a positive attitude toward myself *

*Strongly disagree / Disagree / Agree / Strongly agree*

113. On the whole, I am satisfied with myself *

*Strongly disagree / Disagree / Agree / Strongly agree*

114. I wish I could have more respect for myself *

*Strongly disagree / Disagree / Agree / Strongly agree*

115. I certainly feel useless at times *

*Strongly disagree / Disagree / Agree / Strongly agree*

116. At times I think I am no good at all *

*Strongly disagree / Disagree / Agree / Strongly agree*

**Branch for non-users of Social Media — questions 117 to 146**

*Note.* The questions in this branch are shown only to those who select "I do not use Social Media" in question 11. They replicate the sleep (Pittsburgh) and self-esteem (Rosenberg) blocks, preceded by a question about the reasons for non-use and with an additional item on perceived exclusion (question 136).

117. Why do you not use Social Media? *

*I do not have a phone / I do not like them / I am not allowed to (by my parents, siblings or others) / I prefer to use my phone for other things / I prefer to do other things without my phone / I prefer to be with my friends or my family*

**Sleep quality (Pittsburgh Index) — questions 118 to 135**

118. During the past month, what has usually been your bedtime? *

*9 to 10 / 10 to 11 / 11 to 12 / 12 to 1 / After 1*

119. During the past month, how long has it taken you to fall asleep at night? *

*Up to 15 minutes / 15 to 30 minutes / 30 to 60 minutes / More than 60 minutes*

120. During the past month, what time have you been getting up in the morning? *

*Before 5 / 5 to 6 / 6 to 7 / 7 to 8*

121. How many hours of actual sleep do you estimate you have had each night during the past month? *

*Less than 6 hours / 6 to 7 hours / 7 to 8 hours / 8 to 9 hours / 9 to 10 hours / More than 10 hours*

122. During the past month, how often have you had trouble sleeping… because you could not fall asleep within the first half hour? *

*Not during the past month / Less than once a week / Once or twice a week / Three or more times a week*

123. During the past month, how often have you had trouble sleeping… because you woke up during the night or in the early morning? *

*Not during the past month / Less than once a week / Once or twice a week / Three or more times a week*

124. During the past month, how often have you had trouble sleeping… because you had to get up to use the bathroom? *

*Not during the past month / Less than once a week / Once or twice a week / Three or more times a week*

125. During the past month, how often have you had trouble sleeping… because you could not breathe comfortably? *

*Not during the past month / Less than once a week / Once or twice a week / Three or more times a week*

126. During the past month, how often have you had trouble sleeping… because you coughed or snored loudly? *

*Not during the past month / Less than once a week / Once or twice a week / Three or more times a week*

127. During the past month, how often have you had trouble sleeping… because you felt cold? *

*Not during the past month / Less than once a week / Once or twice a week / Three or more times a week*

128. During the past month, how often have you had trouble sleeping… because you felt too hot? *

*Not during the past month / Less than once a week / Once or twice a week / Three or more times a week*

129. During the past month, how often have you had trouble sleeping… because you had nightmares or bad dreams? *

*Not during the past month / Less than once a week / Once or twice a week / Three or more times a week*

130. During the past month, how often have you had trouble sleeping… because you were in pain? *

*Not during the past month / Less than once a week / Once or twice a week / Three or more times a week*

131. During the past month, how often have you had trouble sleeping… for other reasons? (write them below and indicate their frequency) *

*Open answer.*

132. During the past month, how would you rate your sleep quality overall? *

*Fairly good / Good / Bad / Fairly bad*

133. During the past month, how often have you taken medicine (on your own or prescribed by a doctor) to help you sleep? *

*Not during the past month / Less than once a week / Once or twice a week / Three or more times a week*

134. During the past month, how often have you felt sleepy at school or while doing other activities? *

*Not during the past month / Less than once a week / Once or twice a week / Three or more times a week*

135. During the past month, has it been a problem for you to lack the motivation to do some school or other activities? *

*No problem at all / A very slight problem / A problem / A big problem*

**Self-esteem (Rosenberg Scale) — questions 136 to 146**

136. I feel excluded because I do not use Social Media *

*Strongly disagree / Disagree / Agree / Strongly agree*

137. I feel that I am a person of worth, at least on an equal plane with others *

*Strongly disagree / Disagree / Agree / Strongly agree*

138. All in all, I am inclined to feel that I am a failure *

*Strongly disagree / Disagree / Agree / Strongly agree*

139. I feel that I have a number of good qualities *

*Strongly disagree / Disagree / Agree / Strongly agree*

140. I am able to do things as well as most other people *

*Strongly disagree / Disagree / Agree / Strongly agree*

141. I feel I do not have much to be proud of *

*Strongly disagree / Disagree / Agree / Strongly agree*

142. I take a positive attitude toward myself *

*Strongly disagree / Disagree / Agree / Strongly agree*

143. On the whole, I am satisfied with myself *

*Strongly disagree / Disagree / Agree / Strongly agree*

144. I wish I could have more respect for myself *

*Strongly disagree / Disagree / Agree / Strongly agree*

145. I certainly feel useless at times *

*Strongly disagree / Disagree / Agree / Strongly agree*

146. At times I think I am no good at all *

*Strongly disagree / Disagree / Agree / Strongly agree*

**Depression and anxiety interference (EIDAN) — questions 147 to 156**

*The following questions refer to how you feel. It is important that you answer all the questions below and that you do so honestly, marking the answer you consider appropriate. The questions are presented below:*

*In recent weeks, to what extent has feeling worried, nervous or sad made it…*

147. … difficult for you to get along with your brothers/sisters or other family members? *

*Not at all or hardly at all / A little / Quite a lot / A lot*

148. … hard for you to do your school homework or to study? *

*Not at all or hardly at all / A little / Quite a lot / A lot*

149. … difficult for you to relate to your classmates when you are at school? *

*Not at all or hardly at all / A little / Quite a lot / A lot*

150. … hard for you to do daily activities, such as getting ready for school, eating, sleeping, etc.? *

*Not at all or hardly at all / A little / Quite a lot / A lot*

151. … difficult for you to communicate with your parents, tell them things, etc.? *

*Not at all or hardly at all / A little / Quite a lot / A lot*

152. … so that you do not do as well in exams as you could? *

*Not at all or hardly at all / A little / Quite a lot / A lot*

153. … hard for you, or made you not feel like, meeting up with your friends? *

*Not at all or hardly at all / A little / Quite a lot / A lot*

154. … hard for you to do enjoyable things with your family, such as going to the cinema, going shopping, going to a restaurant, etc.? *

*Not at all or hardly at all / A little / Quite a lot / A lot*

155. … difficult for you to pay attention in class or to ask your teachers questions when you need to? *

*Not at all or hardly at all / A little / Quite a lot / A lot*

156. … difficult for you to make new friends? *

*Not at all or hardly at all / A little / Quite a lot / A lot*